



People's Empowerment Group
ISBM College of Engineering, Nande, Pune
Department of Artificial Intelligence and Machine Learning
Academic Year: 2026 - 27

Project Title:

Sales and marketing agent development for online websites.

Group Members:

1. Barade Kartik Yogiraj
2. Kaname Dnyaneshwari Dattatray
3. Bangar Sayali Hanumant

Problem Definition:

Quality Management System (QMS) consultancy firms face challenges in manually handling technical customer inquiries, qualifying leads, and creating personalized marketing campaigns. Traditional chatbots and rule-based systems cannot accurately understand complex compliance standards such as ISO 9001, IATF 16949, and CE Marking, often resulting in inaccurate or generic responses. This manual process increases response time, workload, and operational costs while reducing lead conversion rates. Therefore, there is a need for an intelligent AI-powered multi-agent system that can provide accurate technical consultation, automatically qualify potential leads, and generate personalized marketing content. The proposed solution uses Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and agent orchestration to automate and optimize the B2B sales and marketing process.

Project Domain:

Artificial Intelligence (AI) and Machine Learning (ML), with a focus on Natural Language Processing (NLP), Large Language Models (LLMs), Multi-Agent AI Systems, and B2B Sales & Marketing Automation.

Literature Survey:

Sr. No.	Title of the Paper	Journal Name, Year	Technique Used	Summary
1.	Large Language Model based Multi-Agents: A Survey of Progress and Challenges	IJCAI 2024	Multi-Agent Systems, LLMs	Reviews the architecture, communication, planning, and collaboration of LLM-based multi-agent systems and discusses their applications and challenges.
2.	Application Level Multi-Agent Methodology for Chatbots	ScienceDirect 2026	LLM, RAG	Describes a production-ready multi-agent chatbot architecture with RAG, specialized agents, and intent routing for customer interaction.
3.	MaRGen: Multi-Agent LLM Approach for Self-Directed Market Research and Analysis	arXiv, 2025	Multi-Agent LLM, Market Research, Task Collaboration	Uses multiple AI agents to automate market research, competitor analysis, and report generation.
4.	AI-Powered Sales and Marketing Analytics Platform Using Multi-Agent AI	IJDIM, 2026	LangGraph, ChromaDB, Multi-Agent	Demonstrates how AI and HR analytics can identify attrition risks early and support employee retention strategies.
5.	AI Salesman: Towards Reliable Large Language Model Driven Telemarketing	AAAI, 2026	LLM, Sales Dialogue, Telemarketing	Introduces a reliable LLM-based telemarketing framework with strategic planning and dialogue optimization.

Scope of the Project:

- Automate customer interaction using an AI-powered sales and marketing chatbot.
- Qualify potential leads automatically based on customer queries and business requirements.
- Generate personalized marketing content for emails and social media campaigns.
- Provide accurate responses using Retrieval-Augmented Generation (RAG) and LLMs.
- Integrate with a CRM system to store and manage customer and lead information.
- Improve sales efficiency by reducing manual pre-sales and marketing efforts.
- Develop a scalable web-based multi-agent AI system for business automation.

Software Requirements:

- **Operating System:** Windows 10/11 or Ubuntu Linux
- **Programming Language:** Python 3.11+, JavaScript (ES6), HTML5, CSS3
- **Development Environment:** Visual Studio Code / PyCharm
- **Frontend Framework:** React.js
- **Backend Framework:** FastAPI (Python)
- **AI Frameworks:** LangChain, LangGraph or CrewAI
- **LLM:** OpenAI GPT-4o / Llama-3 / Mistral
- **Vector Database:** ChromaDB or Pinecone
- **Database:** PostgreSQL or MongoDB
- **Libraries & Frameworks:** Pandas, NumPy, Scikit-learn, Transformers, Sentence Transformers, FAISS/ChromaDB, Streamlit (optional), Uvicorn
- **API Testing:** Postman
- **Version Control:** Git & GitHub

Hardware Requirements:

- **Processor:** Intel Core i5 (11th Gen or higher) / AMD Ryzen 5 or higher
- **RAM:** 8 GB minimum (16 GB recommended)
- **Storage:** 20 GB free SSD space
- **Internet Connection:** Required for LLM APIs, package installation, and cloud database access
- **Graphics (Optional):** NVIDIA GPU (RTX 2050/3050 or higher) for running local open-source LLMs
- **Internet Connection:** Required for package installation and LLM/API integration

Prof. Sangeeta Alagi
Project Guide

Prof. Prajakta Puranik
Project Coordinator

Prof. Kirti Randhe
Head of Department

